

Lecture 5 Partial Differentiations

Multivariable Functions

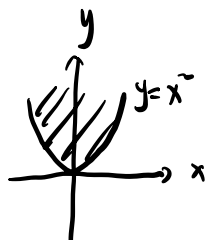


$$V = \pi r^2 h = V(r, h)$$

Domains of functions

$$f(x, y) = \sqrt{y - x^2}$$

$$D = \{(x, y) : y \geq x^2\}$$



$$g(x, y) = \frac{1}{x^2 + y^2}$$

$$D = \{(x, y) : x^2 + y^2 \neq 0\} = \{(x, y) : x \neq 0 \text{ or } y \neq 0\}$$

$$h(x, y) = \frac{1}{xy}$$

$$D = \{(x, y) : xy \neq 0\} = \{(x, y) : x \neq 0 \text{ and } y \neq 0\}$$

Continuous functions.

Def. A function is continuous at (x_0, y_0) if

1) (x_0, y_0) is in the domain of $f(x, y)$

2) for any $\varepsilon > 0$, there exists $\delta > 0$ such that whenever (x, y) is in the domain of f and that $\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$, we have $|f(x, y) - f(x_0, y_0)| < \varepsilon$.

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = f(x_0, y_0)$$

$(x, y) \rightarrow (x_0, y_0)$ Approach (x_0, y_0) for all directions.

Rules.

1) Any polynomial such as $f(x, y) = x^2 + xy + y^2$ is continuous everywhere in $\mathbb{R}^2$

2) $\sin x$, $\cos x$, e^x , $|x|$ are continuous everywhere

3) $\ln x$, $\tan x$, $\sqrt{x}$ are continuous on their domains

4) The $+$, $-$, and $\times$ of two continuous functions are continuous

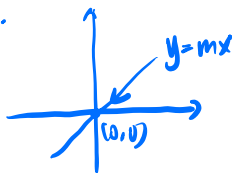
5) The quotient $\frac{f(x, y)}{g(x, y)}$ is continuous at (x_0, y_0) whenever $g(x_0, y_0) \neq 0$

6) The composition $f(g(x, y))$ of two continuous functions $f(t)$ and $g(x, y)$ is continuous.

$$e^{x-y}, \quad \omega, \quad \frac{x}{1+x^2+y^2}, \quad \sqrt{y-x^2}, \quad \frac{1}{xy}$$

Example. Investigate the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{(x+y)^2}{x^2+y^2}$ DNE.

Sol.



$$\lim_{(x,y) \rightarrow (0,0)} \frac{(x+y)^2}{x^2+y^2}$$

if it exists x

$$\times \lim_{\substack{x \rightarrow 0 \\ y=mx}} \frac{(x+mx)^2}{x^2+m^2x^2}$$

$$= \frac{(1+m)^2}{1+m^2} \text{ depends on } m$$

Partial derivatives.

$$f(x) \quad f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

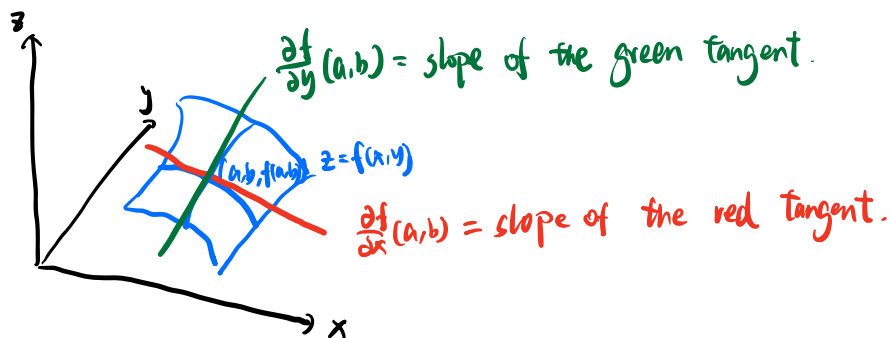
$f(x,y)$

Def. $\frac{\partial f}{\partial x}(x,y)$: = the derivative of $f(x,y)$ with respect to x regarding y constant

$$= \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

$\frac{\partial f}{\partial y}(x,y)$: = the derivative of $f(x,y)$ with respect to y regarding x constant

$$= \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}$$



Notation $\frac{\partial f}{\partial x} = f_x$ $\frac{\partial f}{\partial y} = f_y$ ~~$f'(x,y)$~~

Example. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for the function $f(x,y) = x^2 \sin(xy)$.

$$\text{Sol. } \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x^2 \sin(xy)) = 2x \sin(xy) + x^2 \frac{\partial}{\partial x} (\sin(xy)) \\ = 2x \sin(xy) + x^2 y \cos(xy)$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x^2 \sin(xy)) = x^2 \frac{\partial}{\partial y} (\sin(xy)) = x^2 \cos(xy)$$

Example. Given $f(x,y,z) = e^{x^2+y^3+xy^2}$ compute $\frac{\partial f}{\partial z}$

$$\text{Sol. } \frac{\partial f}{\partial z} = \frac{\partial}{\partial z} e^{x^2+y^3+xy^2} \\ = e^{x^2+y^3+xy^2} \cdot \frac{\partial (x^2+y^3+xy^2)}{\partial z} \\ = xy e^{x^2+y^3+xy^2}$$

Second-order derivatives

Example $f(x,y) = 3x^4y - 2xy + 5y^3$. Compute all first and second partial derivatives,

$$\text{Sol. } \frac{\partial f}{\partial x} = 12x^3y - 2y + 5y^3$$

$$\frac{\partial f}{\partial y} = 3x^4 - 2x + 15xy^2$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (12x^3y - 2y + 5y^3) = 36x^2y$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (12x^3y - 2y + 5y^3) = 12x^3 - 2 + 15y^2$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (3x^4 - 2x + 15xy^2) = 12x^3 - 2 + 15y^2$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (3x^4 - 2x + 15xy^2) = 30xy$$

Notation $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = f_{xx}$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = f_{xy}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = f_{yx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = f_{yy}$$

Theorem (Mixed Partial Theorem). Consider the function $f(x,y)$, if at least one of the second partial f_{xy} and f_{yx} exists and is continuous, then we must have $f_{xy} = f_{yx}$

Example. $f(x,y) = \sqrt{\frac{e^{\sin x}}{x^{2014} + \sqrt{x^{2012} + 1}}} + \cos(xy)$. Find $\frac{\partial^2 f}{\partial y \partial x}$

Sol. $\frac{\partial f}{\partial y} = 0 - x \sin(xy) = -x \sin(xy)$

$$\frac{\partial^2 f}{\partial x \partial y} = - \frac{\partial}{\partial x} (x \sin(xy)) = -\sin(xy) - xy \cos(xy) \text{ is continuous.}$$

$$\Rightarrow \frac{\partial^2 f}{\partial y \partial x} = -\sin(xy) - xy \cos(xy)$$

Def. (C^k functions) A multivariable function $f(x,y)$ is said to be C^0 on

its domain D if it is continuous at any point (x_0, y_0) in D

Moreover, a function $f(x,y)$ is said to be C^k on its domain D if all partial derivatives up to and including order k exist and are continuous at every point (x_0, y_0) in D .

$$f_{xyx} = f_{xxy} = f_{yxx}$$